Vagrant VM Automation tool

- Vagrant is a tool to manage virtual machines automatically. It manages VM lifecycle right from creating VM, making changes and cleaning up VM's.
- Vagrant is not a replacement of any hypervisor like VMware or Oracle Virtual Box rather it works on top of those tools to create and manage virtual machines with more automation.
- It resolves various issues which come when you install VM's manually

VM Management problems which vagrant resolves

- **OS Installation:** There are a lot of steps which we have to go through while installing OS on VM manually
- **Time Consuming:** There are a lot of steps we have to follow, installation also takes lot of time
- **Human error:** It's very natural to make some mistake while following such a hectic process manually
- **Tough to replicate multiple VMs:** When you have to install multiple VM's on different set of machines it's really difficult to follow up as for different systems there might be different ways to follow the same set of action
- **Need to document entire setup:** For following the same installation later for another VM you need to document the entire process

Vagrant for VMs

- **No OS installation:** Vagrant setup VM machines by using VM images, in vagrant they are called as box. These are readymade virtual machines stored on vagrant cloud.
- **Vagrantfile:** All the VM settings will be in a Vagrantfile. You can open this Vagrantfile and edit it if let's say you want to change some settings. We can also do provisioning with Vagrantfile, which means when the OS is completely booted you want to execute some commands on it you can add all these commands also into Vagrantfile and Vagrant will provision it for you.These commands could be related to maybe setting up some server, web server, data base server
- **Some simple Vagrant Commands:**
    - [ ] vagrant init boxname
    - [ ] Vagrant ssh
    - [ ] vagrant up
    - [ ] Vagrant halt
    - [ ] Vagrant destroy

Important Vagrant commands in detail

- **vagrant init** boxname to create a Vagrantfile for that box
- **Vagrant up** command is to start the VM creation process, first time it will perform the booting based on Vagrantfile and start the VM, from next time it will simply start the system without performing the booting
- If later you want to make some changes in the VM just add those changes to your Vagrantfile and then run **vagrant reload**
- Run the **vagrant ssh** to login to the newly created VM
- You can run **vagrant box list** command to see the list of downloaded boxes and **vagrant status** command to check status of VM's inside that folder. If you want to check status of all the vm's use **vagrant global-status**
- **Vagrant halt/destroy** to poweroff or delete the VM respectively

Steps for VM creation using Vagrant:

1. mkdir /e/vagrant-vms mac users can instead creat the directory on desktop using this command mkdir ~/Desktop/vagrant-vms
2. cd /e/vagrant-vms
3. mkdir centos
4. cd centos
5. Search for discover vagrant boxes
6. Search for centos 9
7. Select third option eurolinux-vagrant/centos-stream-9 for ubuntu select ubuntu/jammy64
8. Copy it's name
9. Run command **vagrant init** `eurolinux-vagrant/centos-stream-9`
10. You can check the content by using cat Vagrantfile
11. Run command **vagrant up**
12. You can run **vagrant box list** command to see the list of downloaded boxes and **vagrant status** command to check status of VM.
13. To login to this system you can use **vagrant ssh** command it will use the ssh command in background and login inside the VM